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16 **Research Topic:** The Potential Anti-Inflammatory Effects of Menstrual Blood-Derived Stem Cell Exosomes on Cytokines in Atopic Dermatitis

The Potential Anti-Inflammatory Effects of Menstrual Blood-Derived Stem Cell Exosomes on Cytokines in Atopic Dermatitis

Abstract

Atopic Dermatitis (AD) is a chronic skin disease primarily caused by genetic mutations resulting in a deficiency of the structural protein flaggrin, which is responsible for binding skin cells and creating the skin's natural barrier. Alongside a permeable skin barrier, flaggrin defects also result in a release of epithelial cell-derived cytokines (specifically TSLP, IL-25, and IL-33), responsible for initiating a type 2 immune response inducing inflammation. Current treatments consist of maintenance or reactive therapies, taken frequently and lifelong in order to consistently target these inflammatory cytokines; however, emerging studies have shown a rising potential for the use of menstrual blood-derived stem cells (MenSCs) for immunomodulatory purposes. MenSCs have been known to exhibit potent anti-inflammatory properties; differing from targeted modulation in current treatments, MenSCs can essentially work alongside the immune system, rather than simply blocking receptors, resulting in lasting improvement. This research aims to discover if certain exosomes in the MenSCs secretome can have an anti-inflammatory effect on the cytokines characterized in Atopic Dermatitis through the means of in-sillico experimental methods.

Research Question

Can exosomes in the Menstrual Blood-derived Stem Cells secretome have an anti-inflammatory effect on cytokines characterized in Atopic Dermatitis?

Introduction

Currently, treatments for Atopic Dermatitis consist of various medications that have to be taken frequently, often biweekly and lifelong [1],[2]. The most commonly used treatments include, but are not limited to, topical steroids and biologics (for more moderate to severe cases).[3]. However, since Atopic Dermatitis is a chronic skin disease, this often means that individuals have to use these treatments indefinitely, often relapsing upon discontinuation of these therapies [4]. This raises concerns about certain side effects such as the accumulation of drug toxicity [5] from topical steroids or the loss of effectiveness and localized immune issues (although rare) in biologics [6],[7],[8]. Furthermore, the frequency of biweekly injections causes not only a financial burden for many patients, but a mental burden as well [9],[10],[11]. This brings up the need for research into a longer-term therapy that may be able to accommodate these patients, which brings up a rising research topic at the frontiers of science, holding the potential to address these concerns: Stem Cell Therapies. [12].

Existing Literature

Atopic Dermatitis is a highly complex skin disease that involves not only the integumentary system, but a hyperreactive immune system as well. To understand the hypothetical solution, we must first look at its cause. The disease is known to be primarily caused by the deficiency of the structural protein filaggrin, which is responsible for keeping skin cells cohesive and the creation of the skin's protective barrier. Without filaggrin, the process in which skin cells mature and form this protective layer is disrupted, resulting in a permeable skin barrier. [13]. Another important characteristic of filaggrin defects when discussing Atopic Dermatitis is the release of epithelial cell-derived cytokines like TSLP, IL-25, and IL-33. [14]. These cells cause the secretion of specific cytokines that warn the immune system that there has been an injury or infection, which causes a local inflammatory response to occur. [15].

These released cytokines are the basis for what treatments used today target. Therapies such as biologics or small molecules (commonly found in topical steroids or oral medications) target these specific inflammatory cytokines and either block them (in the case of biologics) or go into the cell to alter gene expression (in the case of small molecules) before they can induce a type 2 immune response. [16]. However, the issue arises in that these therapies are temporary, short-term fixes, as the body eventually breaks down and filters out these molecules or proteins. [17]. This mechanism is what makes the frequency of these therapies so high, and why discontinuation of such treatments will cause relapse in many patients. [18]. Alternatively, rising small-scale studies have introduced a new approach: stem cell therapies.

Stem cells, specifically those found in menstrual blood, have been shown to exhibit extraordinary characteristics that could have the potential to be used as a much longer-term treatment as opposed to existing therapies. Existing literature has found that the MenSCs secretome, or secretion, has significant immunomodulatory and anti-inflammatory properties [19]. MenSCs have been found to secrete the lipid Prostaglandin E2 (PGE2) and the enzyme Indoleamine 2,3-dioxygenase 1 (IDO1), which are biomolecules that can inhibit T-lymphocytes, which are a primary driver in the inflammation caused by Atopic Dermatitis [20],[21]. These are not the only anti-inflammatory markers that MenSCs produce, but rather a quick insight into their potential. Furthermore, MenSCs exhibit immunomodulatory properties as well; for example, we can look at the production of regulatory T-cells (Tregs), which are responsible for suppressing an overreactive immune response. [22]. This describes the fundamental difference in why stem cell therapies could offer longer-term benefit to Atopic Dermatitis patients as opposed to what is used currently, because even when stem cells are filtered out of the body, they leave behind a potent secretome that continues to interact with the immune system [23], [24]. While in vivo studies on MenSCs are limited, small-scale studies have been conducted using their parent cell type, Mesenchymal Stem Cells (MSCs), on patients with Atopic Dermatitis. One such study shows that the injection of Mesenchymal Stem Cells derived from bone marrow has shown

lasting improvements in patients with moderate to severe Atopic Dermatitis, with Eczema Area and Severity Index (EASI) showing improvements of ≥50% after two treatments, and improved symptoms up to an average of 38 weeks [25]. Although studies are limited, especially on MenSCs, we can understand the potential of stem cell therapies using MenSCs, as they exhibit similar properties to bone marrow-derived stem cells, such as low immunogenicity, genetic stability, and anti-inflammatory and immunomodulatory properties. Along with their ease of collection and abundance, this makes them an ideal target to study further [26], [27], [28]. However, existing work currently lacks the link between the anti-inflammatory secretome in MenSCs and its effects on cytokines in Atopic Dermatitis. To address this, this current study serves as a preliminary link to begin the process of uncovering MenSCs' immunomodulatory abilities through the use of various computational models.

Methodology

This study will use a quantitative, in silico multi-stage computational pipeline to first identify our ligand and receptor targets within network pharmacology, then test them against each other via molecular docking, and finally validate our molecular docking data with molecular dynamics. Since this research topic lacks prior biological data in-vivo, we will take a preliminary computational research approach to explore which ligands and receptors to targets work while minimizing ethical, financial and logistical risks by using algorithmic predictions.

Identification of MenSCs Exosomes

Since MenSCs are derived from the endometrium, in their natural state, these molecules actually exhibit pro-inflammatory properties in order to effectively break down and shed the uterine lining. However, when exposed to hyper-inflammatory environments, these Stem Cells release potent anti-inflammatory exosomes to combat it. This means we will be specifically looking for what exosomal cargo components are upregulated during priming.
[27],[29],[30]. This will be carried out through an extensive literature review of peer-reviewed papers; however, since studies are limited, we can also mine public datasets through a multi-omics approach. Since exosomes produce both proteins and microRNA to fight inflammation, we must look at both proteomics (protein) and transcriptomics (microRNA) databases; this hybrid approach is ideal for minimizing error [31],[32]. To standardize these components to Official Gene Symbols, we will use the HUGO Gene Nomenclature Committee (HGNC) database [33].

Identification of Atopic Dermatitis Related Receptors

To identify associated receptors with Atopic Dermatitis, we can look at peer-reviewed literature to see which receptors are known to secrete inflammatory cytokines and are already targeted in

therapies such as biologics or steroids. Furthermore, we can look at bioinformatics databases, namely, GeneCards or DisGeNET,, and search specifically for the terms “Atopic Dermatitis” to find related targets [34],[35],[36],[37].

Network Creation, Filtration and Hub Identification

After both datasets are standardized, we can now look for which proteins show up in both lists; this shows us which molecules associated with Atopic Dermatitis the exosome secretion is able to target. This revised list will then be put into the STRING Database, where it will map a protein-protein (PPI) network; however, we must apply a confidence score filter of ≥ 0.700 to limit false-positive interactions between our datasets [38],[39].

The PPI network created will then be imported into Cytoscape; this is so we can create a visual network map of our protein interactions through the visualization of nodes and lines. To calculate which proteins interact the most with others, we will use the plugin NetworkAnalyzer, which will calculate the Degree Centrality and Betweenness Centrality to tell us how many connections a protein directly has, as well as how often a protein bridges between connections. We will then filter to look specifically at the proteins that reach above the average in both categories, choosing those hub proteins to serve as our candidates for further experimentation [40],[41],[42].

Ligand and Receptor preparation

To prepare our chosen proteins for molecular docking, we will first look at our receptor, the Atopic Dermatitis-associated targets. We’ll begin by downloading high-quality 3D models of these proteins taken from the public database, the Protein Data Bank. However, since these protein models often have water molecules attached to them, we will have to strip any excess chemicals to leave an empty protein ready to dock [43]. For the 3D structure of our ligand, we will optimize and convert using Open Babel, and use MOPAC to calculate exactly how these atoms will interact and behave inside the molecule, ensuring the simulation is accurate [44],[45].

Molecular Docking and Dynamics Validation

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To figure out which molecules have the strongest bonds with our atopic dermatitis-associated proteins, we can use AutoDock to predict how the molecule will bind to the protein; this will not only give us the binding positions but also the binding affinity [46]. We are only looking to keep the strongest-binding molecules. To calculate this, we will use Gibbs free energy; a score of $\leq$ -7.0 kcal/mol is an indicator that a molecule is strongly bonded (we must also check the hydrogen bonding profile to ensure its bonding is for the correct structural reason); therefore, it will be chosen to move forward into molecular dynamics validation [47],[48]. Since molecular docking cannot tell us if the molecule will stay intact under dynamic conditions such as the body,

we must validate our docking findings by using molecular dynamics. We will do this by using the dynamics program OpenMM, placing our best molecule and protein pairs under a configuration simulation that mimics the internal environment of the body [49]. Upon analyzing these results, we can use two distinct metrics to identify if the molecules actually stabilized the protein or not; these metrics are RMSD (to check if the structure stayed stable under dynamic conditions) and RMSF (to check whether or not certain parts of the protein stopped moving as a result of the molecule attachment) [50],[51].

Conclusion

In conclusion, this study serves to bridge the knowledge gap between the complex, hyper-inflammatory chronic skin disease Atopic Dermatitis and the extraordinary properties of menstrual blood-derived stem cells, exploring the potential for a novel, longer-term therapy for Atopic Dermatitis to alleviate the frequency strain of current treatments. Through the use of preliminary computational methods, we can accurately figure out which specific molecules that MenSCs exosomes secrete that could be used in said therapies, while reducing the risk of in-vivo testing on a novel subject area that lacks biological data. However, there are limitations to such a model, as computational methods cannot accurately predict the longevity of these effects if they were to be used as therapies; therefore, further in vitro or in vivo testing needs to take place in order to validate these computational findings. Ultimately, these findings can not only be used as a breakthrough in Atopic Dermatitis therapies, but other hyper-inflammatory diseases as well, as the full potential of MenSCs has yet to be uncovered.

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